Valentin Legras

Senior Data Scientist

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Work Experience

Senior Data Scientist 2020 to now

F. Hoffmann-La Roche, Basel, Switzerland

- Engineered end-to-end PK/PD analysis workflows, from data acquisition and preparation to modeling and reporting, across multiple therapeutic areas (e.g., Oncology, Neuroscience) for Phase 1 to 3 clinical trials.
- Developed interactive data visualization and exploration tools using R/Shiny and Spotfire, enabling real-time data review and supporting cross-functional team decisions, leading to deeper and faster insights from complex clinical data.
- Led the creation of analysis-ready NCA, PK/PD, and ADaM-compliant datasets from raw clinical data for population modeling in R, Phoenix and NONMEM.
- Authored responses to health authority queries (FDA, EMA) and managed the QC and preparation of datasets and reports for electronic regulatory submissions, contributing to the successful progression of molecules from Phase 1 to market, such as COLUMVI(R) (glofitamab) drug.
- Led a process improvement initiative: automation of ADPC, ADNCA, and ADPP datasets by converting legacy SAS programs into reproducible R pipelines, leveraging AI-assisted tools to accelerate development and reduce manual effort by 90%. This new process is now used by 30 data scientists, saving them 50% of their time on these tasks.
- Collaborated with clinical, statistical, and pharmacometric teams to ensure data integrity and deliver custom analysis tools, streamlining modeling and simulation activities.

aNCA Product Owner 2023 to now

F. Hoffmann-La Roche, Basel, Switzerland

- Directed the end-to-end product lifecycle of the 'aNCA' open-source R package using an Agile/Scrum framework, driving it from concept to enterprise-wide adoption by 120+ clinical and pre-clinical scientists at Roche.
- Led, coached, and mentored a core team of 7 developers and managed a 15+ stakeholder network to define a product roadmap now supporting 2,000+ studies annually.
- Delivered more than 1.5M CHF in annual savings by transforming a manual, weeks-long workflow into a 90% automated process. This impact combines 5 FTEs (more than 1M CHF) and more than 500k CHF in retired commercial software licenses.
- Established and enforced GxP-compliant R package development standards, including rigorous code review and mentorship, to ensure regulatory-grade quality.
- Contributed to the pharmaverse ecosystem by developing standardized modules for automated NCA, PK Table, Listing, and Graph (TLG) generation, using Git, CI/CD, Coding Agents (LLMs) and an AI interpretation bot (Python).

IT/Data Science Developer March to August 2020

F. Hoffmann-La Roche, Basel, Switzerland

- Translated pharma scientists' requirements into IT/Data science tasks.
- Deployed several tools with R Shiny (data visualization) to support process modelers and formulators (incl. tidyverse packages).
- Used Git, Jenkins and GCP to build, test and deploy applications.
- Automated database updates on Google Spreadsheet with R (incl. VBA coding), data management and data merging.
- Automated PowerPoint reports with R (officer package) for daily reports.

Data Scientist Oct 2019 to Feb 2020

COFIDIS, Lille, France

- Data cleaning, data analysis, and data science on customer loan data with R.
- Predicted contract return with machine learning methods such as Logistic Regression, Gradient Boosting, and Random Forest.
- Recommended data collection strategies to optimize the rate of returned and completed contracts.

Biostatistical Application Developer May to Aug 2019

Novartis AG, Basel, Switzerland

- Developed an R Shiny application to visualize clinical data (tables, figures and listings) dynamically and interactively during DMC (Data Monitoring Committee) reviews.
- Created dynamic and interactive HTML reports with R Markdown (dplyr, plyr, ggplot, plotly, DT packages).
- Generalized the application to apply across different studies.
- Worked with clinical data, especially ADaM and SDTM datasets (CDISC formats).

SAS Developer June to Aug 2018

Sanofi R&D, Paris, France

- Created a tool to automate data mapping (precise structure) and generated XML files for an artificial intelligence project.
- Developed, validated and referenced SAS programs for dataset creation and reporting: tables, listings and figures.
- Applied SAS/BASE, SAS/MACRO, and SAS/SQL programming.
- Used SDTM/ADaM formats and specifications (metadata files).

SAS and IT Developer — April to Aug 2017

Sanofi R&D, Paris, France

- Adapted SAS programs to a new statistical environment.
- Worked on applications to optimize, improve and develop SAS programs on clinical trials data, enabling biostatisticians to perform their tasks.

Educational Background

Master's Degree, Software Engineering and Statistics — 2017 to 2020

Polytech Lille, Villeneuve-d'Ascq, France

Technical Degree, Statistics and Business Intelligence — 2015 to 2017

Institut Universitaire de Technologie, Vannes, France

Skills

Statistics / Data Science

Statistical programming	■■■■■
Data visualization	■■■■■
Descriptive statistics	■■■■■
Biostatistics	■■■■■

Languages

English (Fluent)	■■■■■
French (Native)	■■■■■
Spanish (B2)	■■■□□

Software

R (Tidyverse, Shiny)	■■■■■
Python	■■■■□
Spotfire / Tableau	■■■■□
SAS	■■■■□

Computing

MS Office (incl. VBA)	■■■■■
Git & Linux	■■■■■
Database / SQL	■■■■□
HTML, CSS, PHP	■■■□□

Publications and Presentations

- Computer-Aided formulation design for pharmaceutical drug product development, part 01: Materials exploration through a visualization tool. Piccione P.M., Lang M.N., Amado Becker F., Hofstetter A., Marchal S., Ly K., Legras V., Ewert A., Kohler D., Maurer R., Willecke N., Burwood R., Kroll P. (2024), International Journal of Pharmaceutics, Volume 667, Part B.
- PBPK Modeling of Entrectinib and Its Active Metabolite to Derive Dose Adjustments in Pediatric Populations Co-Administered with CYP3A4 Inhibitors. Umehara K., Parrott N., Schindler E., Legras V. and Meneses-Lorente G. (2024), Clin Pharmacol Ther, 116: 1130-1140.
- Dexamethasone is associated with reduced frequency and intensity of cytokine release syndrome compared with alternative corticosteroid regimens as premedication for glofitamab in patients with relapsed/refractory large B-cell lymphoma (sunburst plot support).
- Revolutionizing Data Exploration: An Interactive Journey through R Shiny. Legras V., PHUSE 2023.
- R Scripts to Streamline Non-Compartmental Analysis (NCA). Legras V., Suekuer E., PHUSE 2024 (aNCA open-source).
- Sunburst Plots for Analysing the Impact of Drugs on Cytokine Release Syndrome (CRS) Management. Legras V., PAGE 2024. Sunburst visualization method contributed to Hipp J.F. et al., The UBE3A-ATS antisense oligonucleotide rugonersen in children with Angelman syndrome: a phase 1 trial, Nat Med 2025; 31(9):2936-2945.
- Automated QC Framework for Reliable ADNCA Dataset Generation with admiral. Legras V., Aspridis L., PHUSE 2025.

Hobbies

Surfing, Basketball (team), Cycling, Drawing.